1 · Dataset Overview

Dataset Name	Indian Telecom Customer Churn Dataset
Primary File	telecom_churn_cleaned.csv
Raw Source File	raw_data.csv
Records	243,553 rows × 16 columns
Prepared By	Aryan (Data Lead) — SectionC_G5
Last Updated	April 29, 2026
Project Context	Telecom customer churn analysis — Newton School of Technology
Target Variable	churn (0 = Retained, 1 = Churned)
Churn Rate	19.53% (47,562 churned / 243,553 total)

2 · Column Definitions

The cleaned dataset contains 16 columns. Columns 13–16 are engineered features derived during the ETL pipeline.

#	Column Name	Data Type	Description	Sample Values	Notes
1	customer_id	Integer	Unique identifier for each customer record	1, 2, 3, … 243553	Primary key; no duplicates; sequential
2	telecom_partner	Categorical	Telecom operator the customer is subscribed to	Airtel, BSNL, Reliance Jio, Vodafone	4 unique values; near-equal market share (~25% each)
3	gender	Categorical	Customer's self-reported gender	M, F	Binary; M = Male (59.9%), F = Female (40.1%)
4	age	Integer	Customer's age in years at time of record	18, 25, 46, 74	Range: 18–74; mean ≈ 46.1; std ≈ 16.4; no nulls
5	state	Categorical	Indian state where the customer is registered	Karnataka, Maharashtra, Mizoram	28 unique states; near-uniform distribution
6	date_of_registration	Date (String)	Date the customer registered with the operator	2020-01-01, 2022-06-15	Format: YYYY-MM-DD; spans 2020–2023
7	num_dependents	Integer	Number of dependents in the customer's household	0, 1, 2, 3, 4	Range: 0–4; mean ≈ 2.00; std ≈ 1.41
8	estimated_salary	Integer	Customer's estimated annual income in INR	20,000 – 149,999	Mean ≈ ■85,021; std ≈ ■37,509; bounded 20k–150k
9	calls_made	Integer	Average number of calls made per month	0, 44, 62, 108	Range: 0–108; mean ≈ 49.23; std ≈ 29.09
10	sms_sent	Integer	Average number of SMS messages sent per month	0, 24, 45, 53	Range: 0–53; mean ≈ 24.09; std ≈ 14.50
11	data_used	Integer	Average monthly mobile data consumed in MB/GB units	0, 361, 5973, 10991	Range: 0–10,991; mean ≈ 5,010; wide variance (■■ outliers present); std ≈ 2,914
1 2	churn	Binary (Int)	Target variable: whether the customer has churned	0, 1	0 = Retained (80.47%), 1 = Churned (19.53%); no intermediate values

	Column Name	Data Type	Description	Sample Values	Notes
13	registration_year	Integer	Year extracted from date_of_registration — engineered feature	2020, 2021, 2022, 2023	Derived: pd.to_datetime().dt.year; 2023 partial year
14	registration_month	Categorical	Month name extracted from date_of_registration — engineered feature	January, June, December	Derived: dt.month_name(); 12 unique values
15	age_group	Categorical	Derived age segment bucket — engineered feature	Young Adult, Middle Aged, Senior, Elderly	4 buckets: 18–30, 31–45, 46–60, 61+
16	usage_score	Float	Composite engagement score from calls, SMS, and data — engineered feature	2.4, 202.6, 3015.2, 5539.7	Formula: (calls×0.4)+(sms×0.2)+(data×1.0); range: 2.4–5,539.7

3 · Dataset Summary Statistics

Computed directly from telecom_churn_cleaned.csv (243,553 records). All numeric columns have zero null values.

Column	Min	Max	Mean	Std Dev	Nulls	25th %ile	Median	75th %ile
age	18	74	46.08	16.44	0	32	46	60
estimated_salary	20,000	149,999	85,021	37,509	0	51,440	85,094	118,535
calls_made	0	108	49.23	29.09	0	25	49	74
sms_sent	0	53	24.09	14.50	0	12	24	36
data_used	0	10,991	5,009.52	2,914.11	0	2,561	5,028	7,497
num_dependents	0	4	2.00	1.41	0	1	2	3
usage_score	2.4	5,539.7	2,526.86	1,457.11	0	1,266.8	2,515.8	3,768.8
registration_year	2020	2023	2021.20	0.98	0	2020	2021	2022

4 · Categorical Value Distributions

Telecom Partner

Partner	Count	Market Share
Reliance Jio	61,123	25.10%
Airtel	60,905	25.01%
Vodafone	60,802	24.96%
BSNL	60,723	24.93%

Gender

Gender	Count	Share
Male (M)	145,977	59.93%
Female (F)	97,576	40.07%

Age Group

Segment	Count	Share
Elderly (61+)	64,708	26.57%
Senior (46–60)	64,006	26.28%
Middle Aged (31–45)	63,880	26.23%
Young Adult (18–30)	50,959	20.92%

Churn Status

Label	Count	Share
Retained (0)	195,991	80.47%
Churned (1)	47,562	19.53%

Registration Year

Year	Count	Notes
2020	73,126	Full year
2021	72,926	Full year
2022	72,926	Full year
2023	24,575	Partial year (Jan–Apr est.)

5 · Data Quality Assessment

Issue	Status	Detail & Action Taken
Missing_values	None	Zero nulls across all 16 columns. No imputation required.
Duplicate_rows	None	customer_id verified as unique (0 duplicates); 243,553 distinct records.
Outliers — data_used	Present	Range 0–10,991 units with mean ≈ 5,010 and std ≈ 2,914. Wide distribution retained as realistic telecom variance (e.g. heavy mobile-data users).
Outliers — calls_made	Reviewed	Max 108 calls/month; plausible for power users. Retained.
Gender encoding	Consistent	Binary M/F only; no mixed cases or extra whitespace.
Date format	Consistent	All date_of_registration values in YYYY-MM-DD; parseable with pd.to_datetime().
Income range	Reasonable	Bounded ■20,000–■149,999; no implausible values.
Churn encoding	Binary	0/1 only; no intermediate or missing values.
Negative values (raw)	Raw file	raw_data.csv contained a negative data_used value (e.g. row 1: -361). Cleaned in ETL pipeline by taking absolute value.
2023 partial year	Noted	Only 24,575 records for 2023 vs ~73,000 for prior years. Likely represents a partial-year data extract; exclude from year-over-year trend analysis.
Column naming consistency	Consistent	All column names in cleaned file are lowercase with underscores. Aligned with Python/SQL conventions.

6 · Derived / Engineered Columns

The following four columns were created during the ETL/cleaning pipeline and do not appear in the raw source file (raw_data.csv).

Column	Source Column(s)	Data Type	Logic / Formula
registration_year	date_of_registration	Integer	pd.to_datetime(col).dt.year — extracts 4-digit year (2020–2023)
registration_month	date_of_registration	Categorical	pd.to_datetime(col).dt.month_name() — extracts full month name (January … December)
age_group	age	Categorical	18–30 → 'Young Adult'; 31–45 → 'Middle Aged'; 46–60 → 'Senior'; 61+ → 'Elderly'
usage_score	calls_made, sms_sent, data_used	Float	(calls_made × 0.4) + (sms_sent × 0.2) + (data_used × 1.0) — composite engagement index; range 2.4–5,539.7

Note: The reference data dictionary also lists a 'Churn Text' column (Yes/No label derived from churn flag). This column was not present in the cleaned CSV; the binary churn column is the authoritative encoding.

7 · KPI Definitions

KPI Name	Formula	Business Purpose
Churn Rate	(Churned Customers / Total Customers) × 100 → Dataset value: 19.53%	Measures overall customer retention health; primary SLA metric
Retention Rate	(Retained Customers / Total Customers) × 100 → Dataset value: 80.47%	Inverse of churn; used in board-level reporting
Avg Annual Income	mean(estimated_salary) → ■85,021	Proxy for customer lifetime value potential; supports segmentation
Avg Engagement Score	mean(usage_score) → 2,526.86	Indicates product stickiness; correlate with churn likelihood
Operator Market Share	(Operator Customers / Total Customers) × 100	Competitive positioning across Airtel, BSNL, Jio, Vodafone
State Churn Rate	(Churned in State / Total Customers in State) × 100	Identifies geographic risk zones for targeted retention campaigns
Age-Segment Churn Rate	(Churned in Segment / Total in Segment) × 100	Highlights which demographic is most at-risk
Avg Calls (Churned vs Retained)	mean(calls_made) grouped by churn	Behavioural signal — lower usage may precede churn